

Follow the Workers

Do labor-market flows predict which U.S. industries outperform?

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1 Executive summary

We test whether public labor-market flows—job openings, hires, quits, layoffs, hours and wages from the BLS JOLTS and CES surveys, measured *as they were known* on each decision date—predict which of 13 U.S. industry groups outperform over the next one to three months, and whether labor-market tightness changes the answer. A preregistered, market-neutral industry rotation was frozen on research data (2004–2014) and evaluated once on 142 untouched months (November 2014–August 2026). It lost money: the primary strategy returned -4.6% a year after costs (Sharpe -0.58), its forecasts were uncorrelated with realised returns, and a placebo test cannot tell it apart from noise ($p = 0.72$). **We conclude: Do not implement.** The failure is informative. (i) The prespecified signs were wrong for two of four signals, and the one labor variable with a stable link to returns—*wage growth*, rank IC $+0.04$ in both the research and the test period—had been dropped during the build; restored post hoc, a plain wage-growth tilt earns a Sharpe of about **0.25** in both periods, which we freeze for a live forward test rather than claim as alpha. (ii) Hiring behaves as investment-based asset pricing predicts at a 12-month horizon (more hiring, lower subsequent returns), not at the one-month horizon we traded. (iii) Our research sample never contained a tight labor market ($V/U > 1$ in 0 research months versus 75 test months), so the tightness hypothesis could not be tested as designed. (iv) In a blinded experiment, LLM agents searched for new labor signals: a more capable model followed the protocol far better but found no more out-of-sample signal, and every agent feature that passed research failed the holdout.

[TODO: one sentence on what Claude was good and bad at as a research assistant]

2 What did we try?

2.1 Idea: labor flows as early, public, industry-level news

Firms reveal their plans in the labor market before they reveal them in earnings. Hiring, quitting and layoffs are published monthly by industry, are free, and are rarely used in industry-rotation strategies, which typically rely on price momentum (Moskowitz and Grinblatt, 1999; Zarattini and Antonacci, 2024). Two economic channels predict *opposite* signs, which is what makes the question interesting (Figure 1):

- **Demand news** (short horizon, positive sign). Rising openings, hires and hours signal growing demand for an industry’s output before it reaches reported sales.
- **Cost and investment news** (longer horizon, negative sign). In investment-based models, firms that hire heavily are converting growth options into assets in place and face higher adjustment costs, so their expected returns fall (Belo et al., 2014). Wage pressure squeezes

margins. Conversely, labor-market frictions make labor-intensive firms riskier and raise their required returns (Kuehn et al., 2017; Donangelo et al., 2019).

- **Tightness as moderator.** When vacancies outnumber the unemployed, hiring is expensive and wage news should dominate demand news. We measure tightness by $T_t = \log(\text{job openings}_t / \text{unemployed}_t)$.

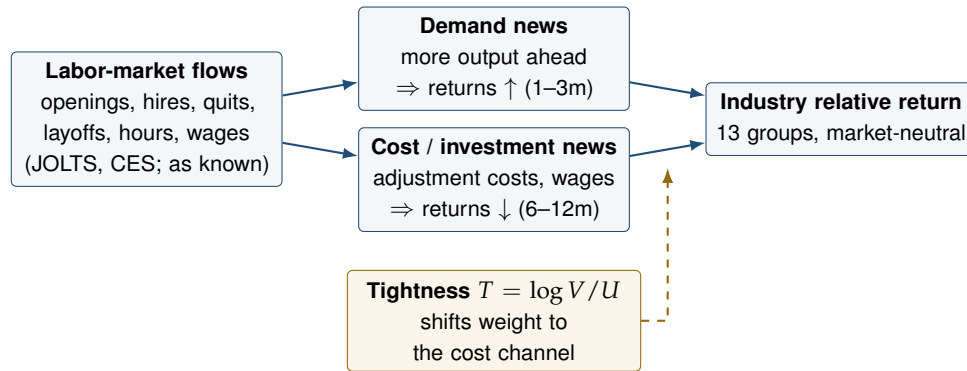


Figure 1: Thesis. Labor flows carry two kinds of industry news with opposite sign and different horizons; tightness should change which one dominates.

Table 1 lists each signal, the sign we preregistered and the mechanism behind it. The last column, added after the fact, shows what the data say; we discuss it in Section 3.3.

Table 1: Signals, preregistered signs and what the data say POST-HOC (last column only)

Signal	Definition (industry g , month t)	Prior	Mechanism	IC, $h=1$
F1 Openings	12-month change in 3-month-avg. openings rate	+	demand news	−0.005
F2 Hires	12-month change in hires rate	+	demand news; investment channel: − at long horizon	+0.013
F3 Quits	12-month change in quits rate	none	worker bargaining power	+0.009
F4 Layoffs	12-month change in layoffs rate	−	negative demand shock	+0.028
F5 Hours	12-month log change, avg. weekly hours	+	intensive-margin demand	−0.012
W Wages	12-month log change, avg. hourly earnings	dropped	cost pressure (−) vs. labor risk premium (+)	+0.040

Rank IC = average monthly Spearman correlation between the signal and next-month industry relative return, May 2004–Aug 2026, revised data with publication lags. Only W has $|t| > 2$ (2.4). The fixed rule A0 used $F1+F2-F4+F5$; the data reject the signs of F4 and F5.

2.2 Data

Universe. The 49 French industries are mapped to the 13 JOLTS/CES industry groups and value-weighted within each group (Appendix C). The mapping is coarse: labor data describe all establishments in an industry, while returns describe listed firms only. We return to this in Section 3.3.

Point-in-time data. Labor data are revised for months after first release. Using today’s revised values would leak information. We rebuilt every series as it stood on each decision date from

Table 2: Data sources. All free; links in Appendix B.

Source	Series	Freq.	Known at t	Vintage handling
BLS JOLTS	openings, hires, quits, layoffs rates; 13 industries, not seasonally adjusted	M	$t - 2$	ALFRED vintages: value as known on each decision date
BLS CES	avg. weekly hours, avg. hourly earnings (production & nonsupervisory)	M	$t - 1$	as above
FRED	total openings (JTSJOL), unemployed (UNEMPLOY) for T_t	M	$t - 2$	as above
Ken French	49 value-weighted industry portfolios $\rightarrow$ 13 groups; FF5 and momentum factors	M	t	CRSP build 202608

ALFRED vintages (78 series), and kept three alternatives—first release, fully revised, and revised with fixed publication lags—to measure how much revisions matter (Appendix D).

Sample. Research (signal design and model selection): May 2004–Sep 2014, 125 months. Test, opened once: Nov 2014–Aug 2026, 142 months. [TODO: Figure: timeline of T_t with research/test shading and the $V/U = 1$ line (fig_timeline.pdf)]

2.3 Implementation

Signals. For a labor series $x_{g,t}$ in industry group g , let $\bar{x}_{g,t} = \frac{1}{3}(x_{g,t} + x_{g,t-1} + x_{g,t-2})$. With publication lag ℓ (2 months for JOLTS, 1 for CES),

$$F_{g,t} = \bar{x}_{g,t-\ell} - \bar{x}_{g,t-\ell-12} \quad (\text{log differences for hours and earnings}). \quad (1)$$

Each signal is scaled by its own expanding-window volatility (at least 24 months), then z-scored across the 13 groups and winsorised at ± 3 , giving $z_{g,t}^{(k)}$.

Models. Table 3 summarises the five preregistered strategies. Learned models are ridge regressions of next-month relative return on the signals,

$$\hat{\beta}_t = \arg \min_{\beta} \sum_{s < t} \sum_g (r_{g,s+1}^{\text{rel}} - z_{g,s}^{\top} \beta)^2 + \alpha \|\beta\|^2, \quad (2)$$

re-estimated each month on an expanding window, with $\alpha \in [0.01, 1000]$ chosen by blocked cross-validation inside the research period.

Table 3: Preregistered strategies

A0	Fixed rule, no estimation: $S_{g,t} = \frac{1}{4}(z^{(1)} + z^{(2)} - z^{(4)} + z^{(5)})$.
A1	Ridge on F1–F5.
A1-T	A1 plus interactions of each signal with tightness terciles.
A3	A1 plus the two best features proposed by <i>independent</i> LLM agents (Section 2.3).
A4	A1 plus the two best features proposed by <i>communicating</i> LLM agents.

Primary strategy chosen on research data only: A3.

Portfolio, risk model and costs. Each month the score is converted to dollar-neutral industry weights $\tilde{w}_{g,t}$ (long the highest-ranked groups, short the lowest). Because the forecast is for one month but labor news is slow, we hold each book for three months in overlapping tranches, $w_{g,t} = \frac{1}{3} \sum_{j=0}^2 \tilde{w}_{g,t-j}$. The book is hedged to zero market beta (60-month betas) and scaled to a 10% ex-ante volatility using a 60-month covariance matrix, subject to gross exposure ≤ 2 . Costs are 10bp per unit of industry turnover plus 2bp for the hedge; results at 0 and 25bp are in Appendix D. [TODO: verify the weight formula against backtest() in the frozen code and write it as an equation]

How we kept the test honest

- Every input as it was known on the decision date (ALFRED vintages).
- Signals, models, primary strategy and pass/fail gates fixed and hashed before any test data were touched; the test period was opened exactly once.
- Every result reported whatever its sign; a circular-shift placebo with full refits, drop-one-industry checks, Holm-adjusted bootstrap comparisons and a deflated Sharpe ratio (Bailey and López de Prado, 2014).
- Anything decided after seeing test results is labelled **POST-HOC**; the live out-of-sample book is labelled **FORWARD TEST**.

AI-assisted signal search. We asked LLM agents to propose new labor features from six anonymised variables (V01–V06) using a restricted formula grammar. Two arms of three agents each—*independent* versus *islands* that exchange short migration reports—ran three repetitions, for 108 proposals per study. Selection was mechanical (research IC). An LLM judge, audited by a human, scored whether ideas spread between agents. [TODO: Figure: agent design (fig_agents.pdf), one line per box]

Claude as research assistant. We used Claude (Anthropic) at four points of the build: reviewing sources and code, proposing signals as blinded agents, judging whether ideas spread between agents, and red-teaming the robustness plan. We used it again to review the finished study. Three of the prompts are below; outputs and our evaluation are in Section 3.4 and transcripts in Appendix A.

Prompt 1 — Source and code review (P1)

[TODO: short exact excerpt from outputs/p1_request.json]

Prompt 2 — Blinded signal-search agent (P2)

[TODO: short exact excerpt of the agent prompt template (params.py)]

Prompt 3 — Robustness red team (P4)

[TODO: short exact excerpt from outputs/p4_request.json]

3 What did we learn?

Table 4 reports the one-time test. No strategy made money; the primary lost 4.6% a year with a 41% drawdown and failed four of five preregistered gates. Costs are not the explanation: they run at 1.0–1.35% a year against gross losses of 2.5–4.8%.

Table 4: Preregistered strategies: research versus one-time test**CONFIRMATORY**

Strategy	Research	Test, Nov 2014–Aug 2026 (142 months)			
	Sharpe	Sharpe	Net return	Volatility	Max drawdown
A0 fixed rule	−0.44	−0.36	−3.5%	9.7%	−40.3%
A1 ridge	−0.54	−0.51	−4.0%	7.9%	−37.7%
A1-T tightness	−0.55	−0.59	−4.9%	8.2%	−36.3%
A3 primary	−0.13	−0.58	−4.6%	7.8%	−41.1%
A4 islands	−0.18	−0.79	−6.1%	7.8%	−53.3%

Annualised, net of costs, point-in-time data. Primary: IC -0.021 ($t = -1.08$); placebo $p = 0.72$; break-even cost -32 bp. Gates passed: 1 of 5. Verdict: Do not implement.

3.1 Style and risk exposures

Regressing the primary’s monthly return on the Fama–French five factors, momentum and industry momentum: the “labor” book is partly a quality-and-growth tilt, loading on profitability (RMW $+0.19$, $t = 2.2$) and against value (HML -0.15 , $t = -2.0$), with no market or momentum exposure and an alpha of -0.38% a month ($t = -1.86$). [TODO: factor-loading figure for A3 and the W book (fig_loadings.pdf); one sentence on why labor signals pick up RMW]

3.2 Performance over time

[TODO: Table: A3 and W book over full sample / post-2010 / last 18 months. Known for A3: first half -0.72 , second half -0.49 , ex-pandemic -0.51 , last 18 months $+0.85$ (src results.json windows.A3). Figure: cumulative net returns with industry-momentum benchmark and the W book dashed and tagged POST-HOC (fig_cumulative.pdf).]

3.3 Robustness and limits: why it failed

Wrong signs, missing signal. Single-signal ICs (Table 1) show layoffs and hours enter A0 with the wrong sign, and openings carry no information. Wage growth, which the original blueprint included and the build dropped, is the only signal with a consistent IC: $+0.039$ in research and $+0.041$ in test. Both agent runs independently rediscovered it from the anonymised variable V05 without being told what it was.

Wrong horizon. At one month no hiring signal predicts returns; at twelve months hires (-0.062 , $t = -2.1$) and quits (-0.065 , $t = -1.7$) carry the negative sign the investment channel predicts (Belo et al., 2014). We traded the demand channel at a horizon where only the cost channel shows up. [TODO: IC-by-horizon heatmap (fig_horizon.pdf)]

An untestable moderator. Tightness terciles were set on an expanding window. Because the research period never saw a tight market, 132 of 141 test months fell into the “high” tercile, so the tightness model was effectively a single regime. A balanced alternative— T rising versus falling over 12 months—splits the test 88/53. [TODO: IC in T-rising vs T-falling months (fig_tightness.pdf), labelled POST-HOC]

Fragile agent features. The selected A3 features were gated on the “low tightness” regime and were active in only 3 of 142 test months. Every feature that passed research failed the test (4/4 Sonnet, 6/6 Opus). Regime conditioning produced the highest research t -statistics in the whole

search (3.56) and nothing out of sample. [TODO: scatter research IC vs test IC, selected and regime-gated features marked (fig_agents_scatter.pdf)]

Concentrated losses. Health care alone cost the primary 0.26 of cumulative return; construction and finance followed. [TODO: P&L by industry bar chart (fig_industry_pnl.pdf)]

Over-shrinkage and a binding risk cap. Cross-validation chose the largest ridge penalty on the grid ($\alpha = 1000$) for every model, and the gross-exposure cap, not the volatility target, bound in 80–93% of months.

Limits. 142 test months can only detect a Sharpe of about 0.6 at $t \approx 2$; a null does not rule out a smaller effect. The 49-to-13 industry mapping dilutes the labor signal.

What survives **POST-HOC**. [TODO: W-tilt book, specification frozen in analysis/v2_spec.md before running: Sharpe by window, factor alpha, deflated Sharpe counting ≈ 108 looks, placebo, costs 0/10/25 bp. Quick check (3 Oct): Sharpe ≈ 0.36 research, 0.23 test, 0.28 full, negative last 18 months. Report whatever the frozen run gives.]

Live forward test **FORWARD TEST**. [TODO: v2 specification hashed on 3 Oct 2026; the Q4 2026 book (decision 30 Sep 2026, held Oct–Dec) and its ETF-proxy tracker.]

3.4 Claude evaluation

[TODO: Table: interaction | what we asked | what it got right | what it got wrong | what we changed | our critique. Rows P1–P4 (build), P5 (post-mortem review), P6 (this iteration). Critique cells are written by team members.]

Agents: capability is not alpha. [TODO: tighten] Compared with Sonnet, Opus produced more valid proposals (106/108 vs 98/108), more admitted migration reports (15/18 vs 7/18) and far more usable judge labels (84/88 vs 8/36), but no feature that survived the test. Communication raised research search scores for one model (best IC 0.136 vs 0.059) and lowered them for the other (0.049 vs 0.083); the gap never exceeded the seed-to-seed spread. The LLM judge agreed with the human on 10 of 17 idea-uptake calls, and all 7 disagreements were the machine claiming an influence the human did not see.

Reflection. [TODO: strengths and limits of Claude as an investment research assistant; what we would do differently]

References

- David H. Bailey and Marcos López de Prado. The deflated sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.
- Frederico Belo, Xiaoji Lin, and Santiago Bazdresch. Labor hiring, investment, and stock return predictability in the cross section. *Journal of Political Economy*, 122(1):129–177, 2014.
- Andres Donangelo, François Gourio, Matthias Kehrig, and Miguel Palacios. The cross-section of labor leverage and equity returns. *Journal of Financial Economics*, 132(2):497–518, 2019.
- Lars-Alexander Kuehn, Mikhail Simutin, and Jessie Jiaxu Wang. A labor capital asset pricing model. *Journal of Finance*, 72(5):2295–2332, 2017.

Tobias J. Moskowitz and Mark Grinblatt. Do industries explain momentum? *Journal of Finance*, 54 (4):1249–1290, 1999.

Carlo Zarattini and Gary Antonacci. A century of profitable industry trends. SSRN Working Paper 4857230, 2024.

A Claude interaction transcripts

[TODO: condensed transcripts for P1–P6 with links to the files]

B Data links

[TODO: one line per source with URL]

C Industry mapping

Group	Ken French industries
Mining	Gold, Mines, Coal, Oil
Construction	Cnstr
Durable mfg	Toys, BldMt, Steel, FabPr, Mach, ElcEq, Autos, Aero, Ships, Guns, Hardw, Chips, LabEq, MedEq
Nondurable mfg	Food, Soda, Beer, Smoke, Hshld, Clths, Drugs, Chems, Rubbr, Txtls, Paper
Wholesale	Whlsl
Retail	Rtail
Transport & utilities	Trans, Util
Information	Telcm
Finance & insurance	Banks, Insur, Fin
Real estate	RIEst
Professional & business services	BusSv
Health care	Hlth
Accommodation & food	Meals

D Full results

[TODO: vintage variants (as-known / first / revised / fixed-lag), cost sensitivity, gate table]

E Code snippets

[TODO: as-of lookup, signal construction, backtest core (15–25 lines each)]

F Follow-up study with a stronger model

[TODO: Opus follow-up: launched after the test was opened, so exploratory. Primary A4 Sharpe +0.01, driven by a long position in durable manufacturing (semiconductors).]

G Forward-test specification

[TODO: frozen spec, hash, timestamp, Q4 2026 book]